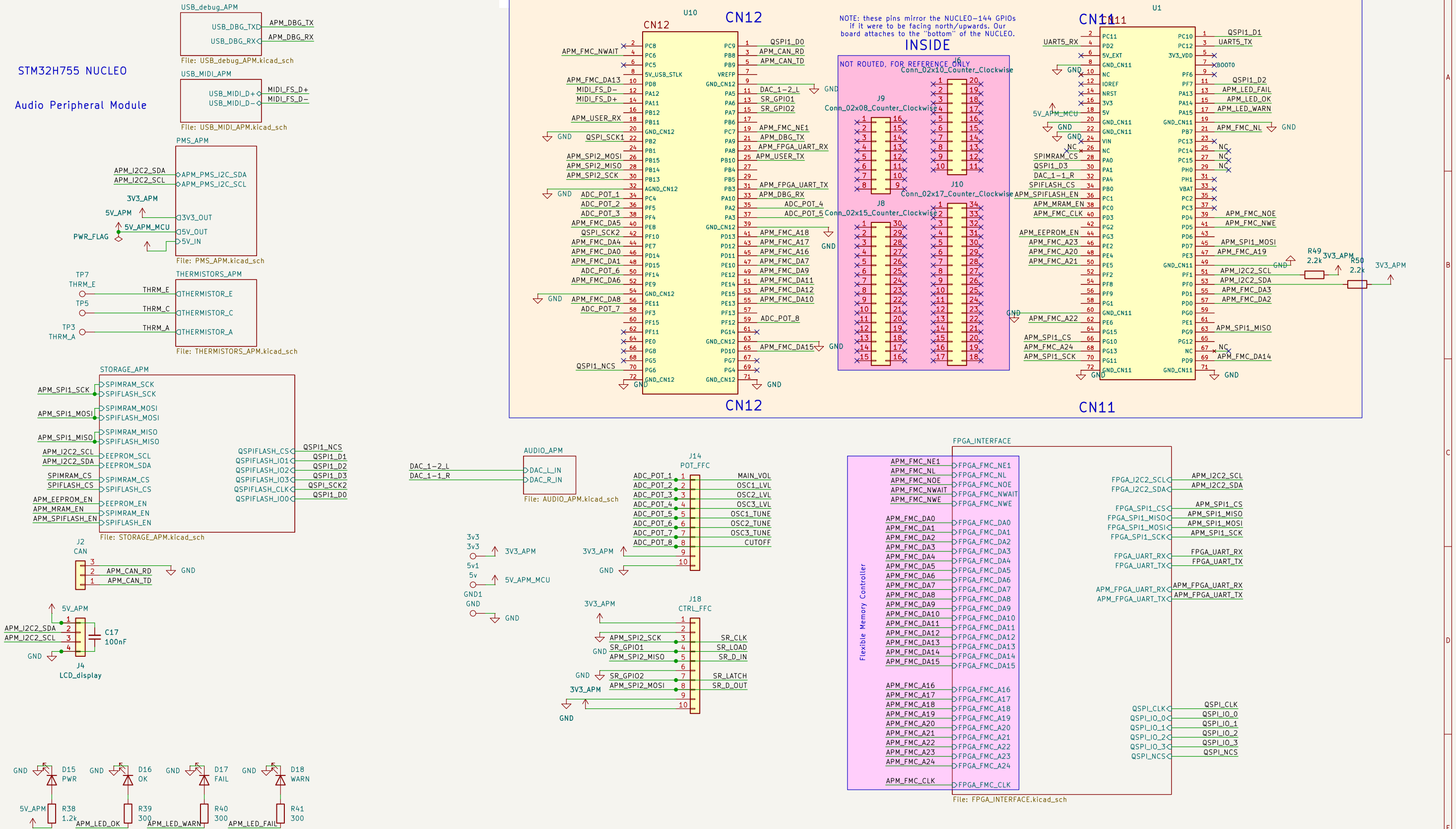


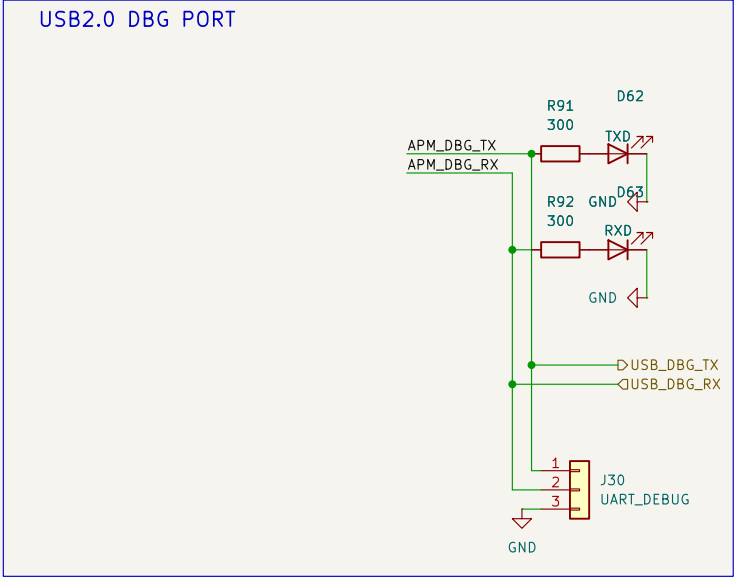
STM32H755 NUCLEO

Audio Peripheral Module



UART -> USB Debug Port

UART debug messages available via a USB port and a 3 pin jumper for TX, RX, and GND UART connections



Sheet: /USB_debug_APM/
File: USB_debug_APM.kicad_sch

Title:

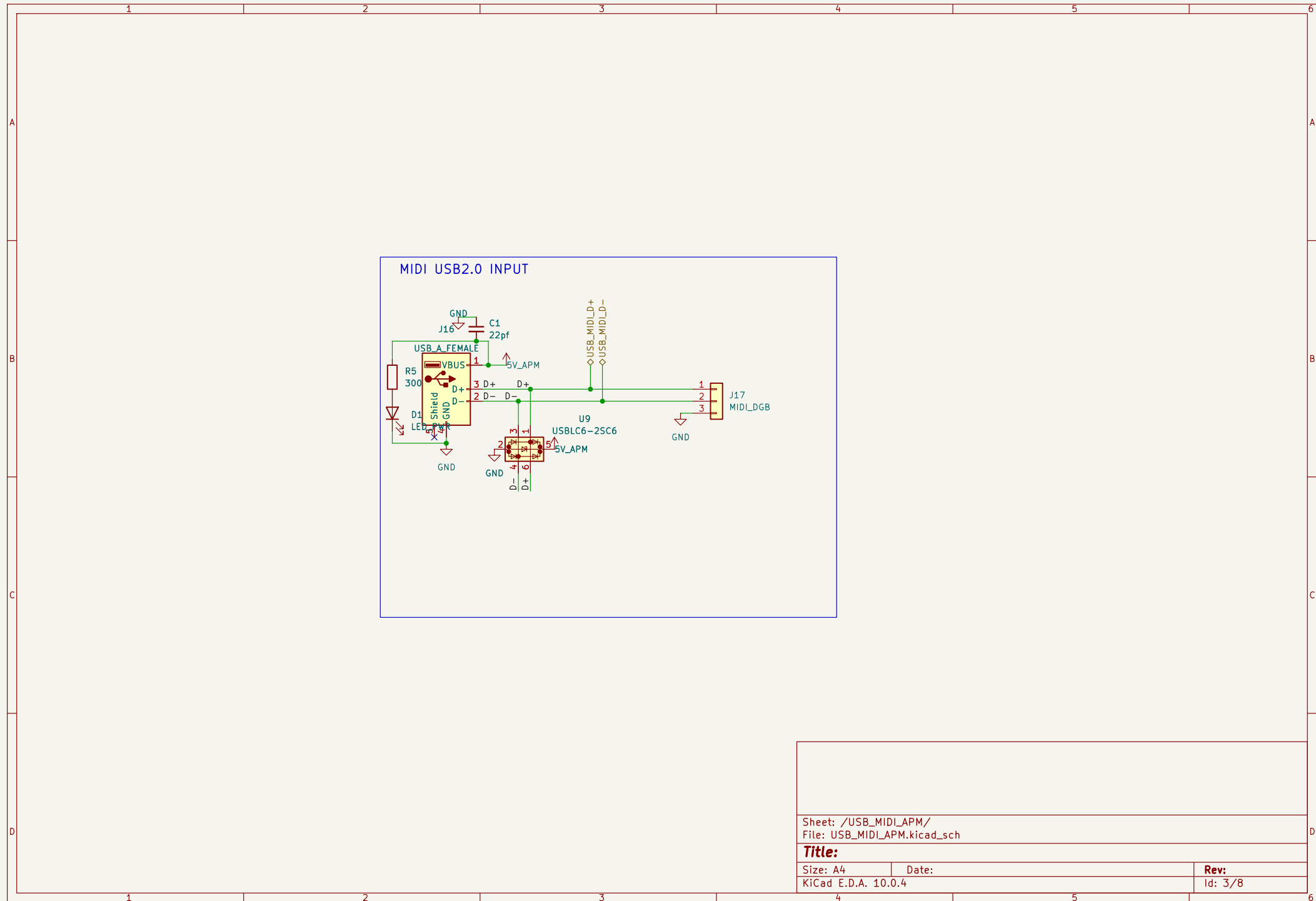
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KiCad E.D.A. 10.0.4

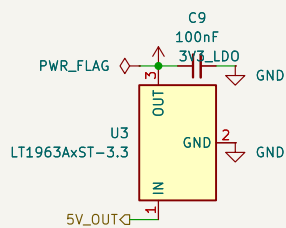
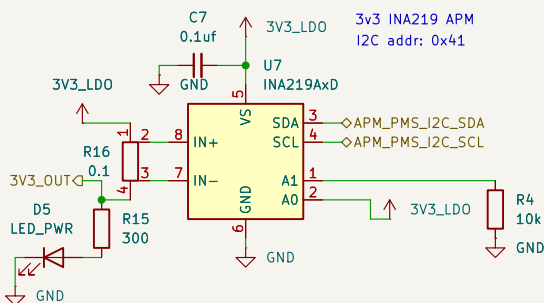
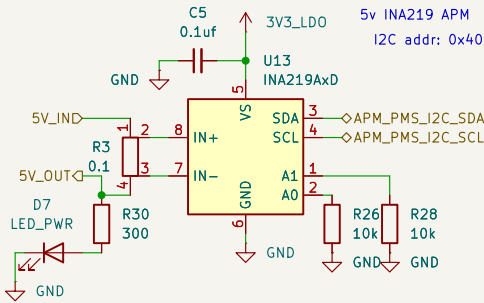
Rev:

Id: 2/8



Power Monitor and Supply

PMS has an INA219 current sensor
for each board/node in the
PCB. Each takes in the 5V from it's
parent board and supplies 5V to said
board's circuit (its peripherals)
thus enabling monitoring



Sheet: /PMS_APM/
File: PMS_APM.kicad_sch

Title:

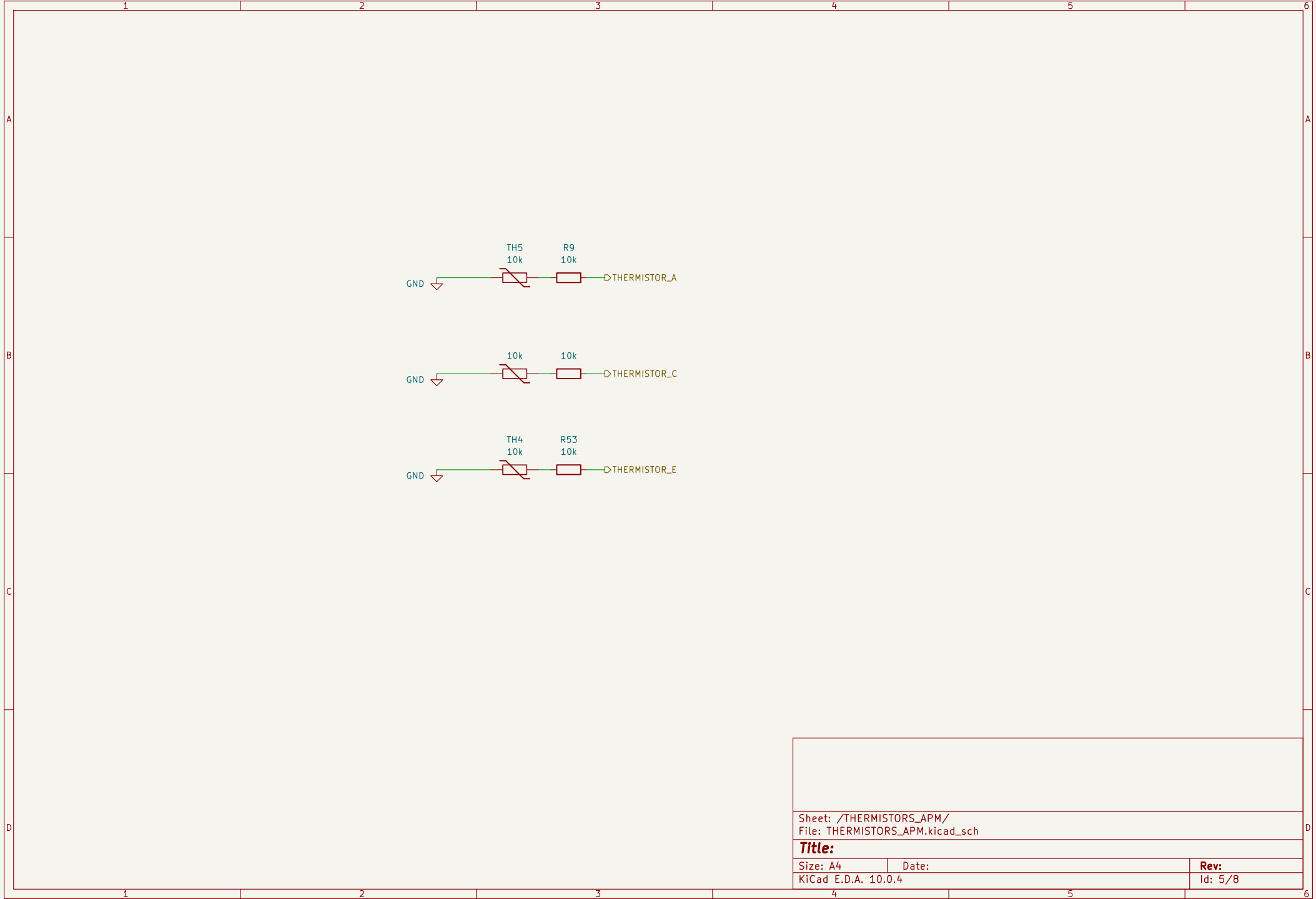
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KiCad E.D.A. 10.0.4

Rev:

Id: 4/8



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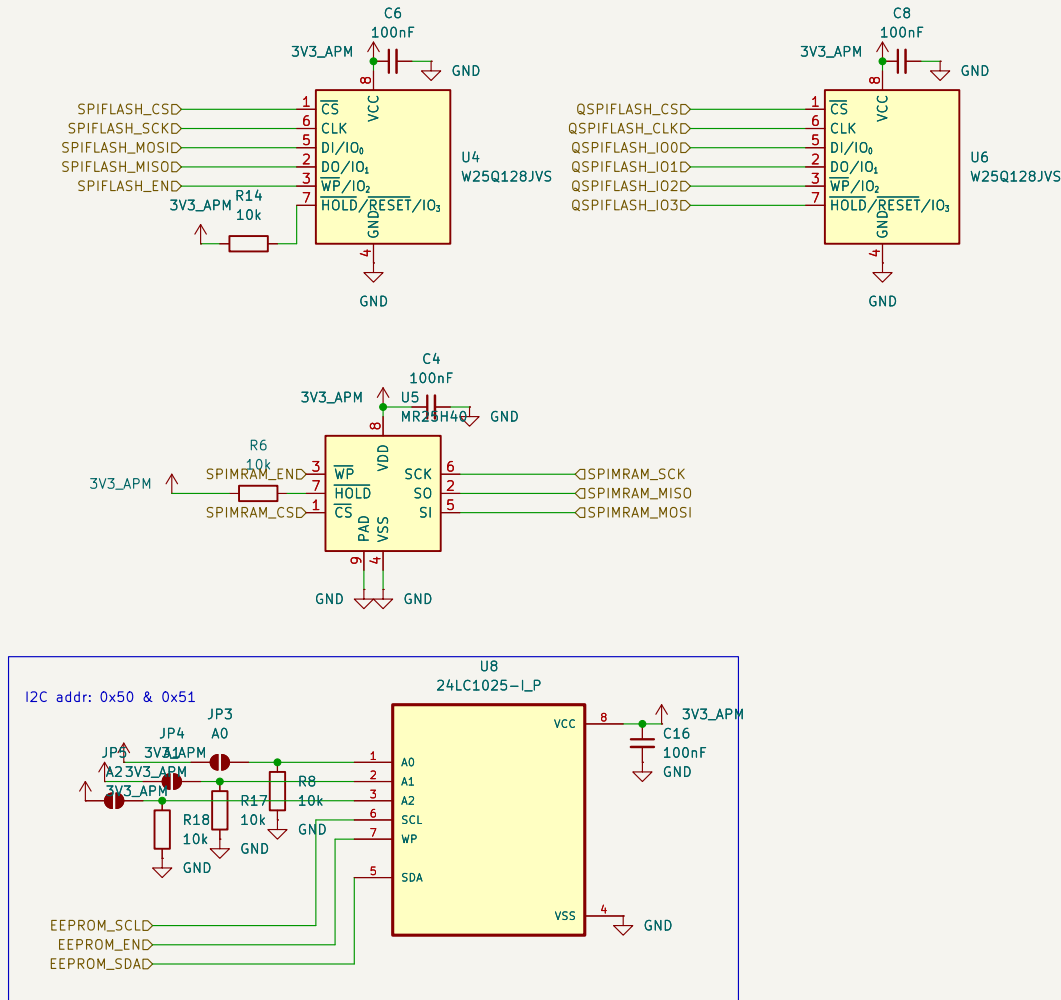
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Date:

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Rev:

Id: 5/8



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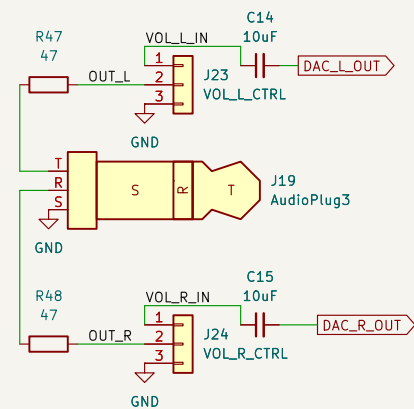
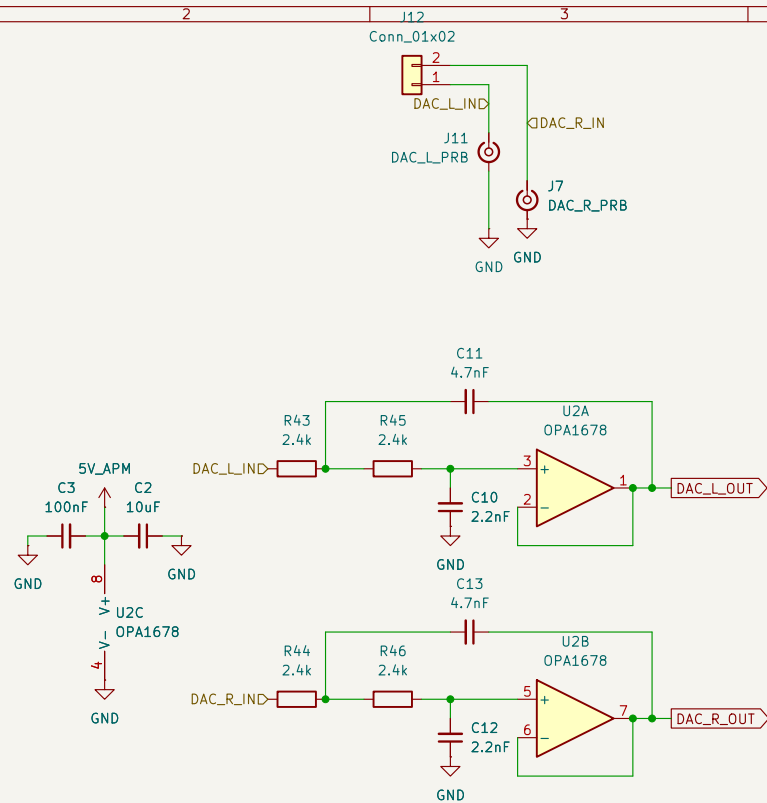
Size: A4

Date:

Rev:

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Id: 6/8



Sheet: /AUDIO_APM/
File: AUDIO_APM.kicad_sch

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Size: A4

Date:

KiCad E.D.A. 10.0.4

Rev:

Id: 7/8

ACM FPGA Interface

Dual FPC STM32<->Tang Nano 20K FPGA connection configuration. The STM32 side has two connectors where only one can be active at a time

- Configuration 1: FPGA_FFC with SPI, QSPI, UART connections.
- Configuration 2: FPGA_FMC with:
 - 16 bit muxed data
 - and async FMCconfiguration resulting in a 64MB shared address space between the STM32 & FPGA

